

UNIVERSITY OF CALIFORNIA, DAVIS
Department of Mechanical & Aerospace Engineering

EME-185A

Winter 2026

Report #2 Preliminary Machine Architecture Report

NAMES _____ GRADE _____/100

I. GENERAL CONSIDERATIONS _____/20

- 1.1) Report format _____/5.0
Appropriate length, figure/tables in text and referenced, use of appendixes appropriate, etc.
1.2) Overall document presentation _____/5.0
Grammar, vocabulary, appropriate technical references, professional in appearance and content, etc.
1.3) Table of contents included _____/5.0
Accurate page numbers, headings and sub-headings used
1.4) Readable by a non-expert _____/5.0
Define all acronyms, defines domain-specific jargon, correct usage of units/engineering terms, etc.

II. ABSTRACT / EXECUTIVE SUMMARY _____/10

- 2.1) Concise, clear, focused, easy to understand, technically accurate

III. SYSTEM ARCHITECTURE SELECTED _____/20

- 3.1) Statement of selection criteria used to design system _____/5.0
Well defined requirements, design and requirements related, subsystem selection is clear
3.2) Layout drawings/figures of system and subsystems _____/7.5
Works with text to describe function, visuals elements to show function (arrows, expld views, etc.)
3.3) Text description of system solution selected _____/7.5
Clearly describes every aspect of how the design works

IV. DISCUSSION _____/25

- 4.1) Discussion of requirements and rolldown to subsystems _____/5.0
Half points well described requirement updates, half points roll down discussion
4.2) Discussion of various subsystems and their interface _____/10.0
Half marks accurate and appropriate technical discussions, describes how subsystems work together
Half marks for proper engineering drawing package
4.3) Proof that design will meet requirements (eg. calculations) _____/10.0
Accurate, well motivated/ related to requirements, reasonable assumptions stated

V. PRELIMINARY BILL OF MATERIALS & SIMPLE COST ESTIMATE _____/10

- 5.1) All parts included, each part belongs to a sub system, accurate costs, all info included

VI. PROJECT MANAGEMENT INFORMATION

6.1) Appropriate tasks identified over 2 quarters

 /5.0

6.2) Updated Gantt chart for 2 quarters

 /5.0

6.3) Critical path method analysis

 /5.0

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